Stomatal traits across cytotypes — short summary

Automated analysis of microscope images, biophysical traits, and cytotype comparison

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Summary

We automatically detected and measured stomata in **3,253 microscope images** from **540 plants** (5 populations; four cytotypes; both leaf sides), yielding **104,278 detected stomata**. Key biological results:

- **Cell size follows ploidy, but not linearly.** Synthetic neotetraploids (`neo_4x`) exhibit a pronounced "gigas" effect with the largest stomata (23.9 μm abaxial, +27% vs. `nat_2x`), while established natural tetraploids (`nat_4x`) display secondary evolutionary downsizing to intermediate cell size (20.6 μm, +10%). Diploids `ory_2x` and `nat_2x`: 18.6 and 18.8 μm.
- **Density & size–density trade-off.** Abaxial density is 253 mm^{-2} in `nat_2x`, 235 in `nat_4x` (-7%, not significant), 162 in `neo_4x` (-36%, $p < 0.001$) and 295 in `ory_2x`.
- **Anatomical conductance ($g_{s,\max}$) & SPI.** Abaxial $g_{s,\max}$ (mol m^{-2} s^{-1}) under standardized maximum pore aperture: `neo_4x` 0.60, `nat_4x` 0.73, `nat_2x` 0.73, `ory_2x` 0.83. While `neo_4x` suffers a severe -24% drop in diffusion capacity, `nat_4x` compensates through packing to restore $g_{s,\max}$ to diploid levels. The aperture-free Stomatal Pore Area Index (SPI) mirrors this pattern (Table 1).
- **Both leaf surfaces bear stomata.** Rather than having stomata restricted to the lower surface, all cytotypes develop stomata on both the lower (abaxial) and upper (adaxial) leaf surfaces. Stomatal density is consistently ~20–25% higher on the lower surface across all cytotypes.

(Click any highlighted term to jump directly to its explanation in the Glossary & Technical Definitions.)

Methods in brief

Image analysis & trait extraction. A YOLO11 instance-segmentation model trained on manually annotated crops plus hard-negative background tiles was executed across full-resolution microscope images. For each stoma, we extract full stoma dimensions, guard-cell thickness and pore dimensions; field density is computed per calibrated image area (Figure 1). Size statistics exclude edge and out-of-focus stomata.

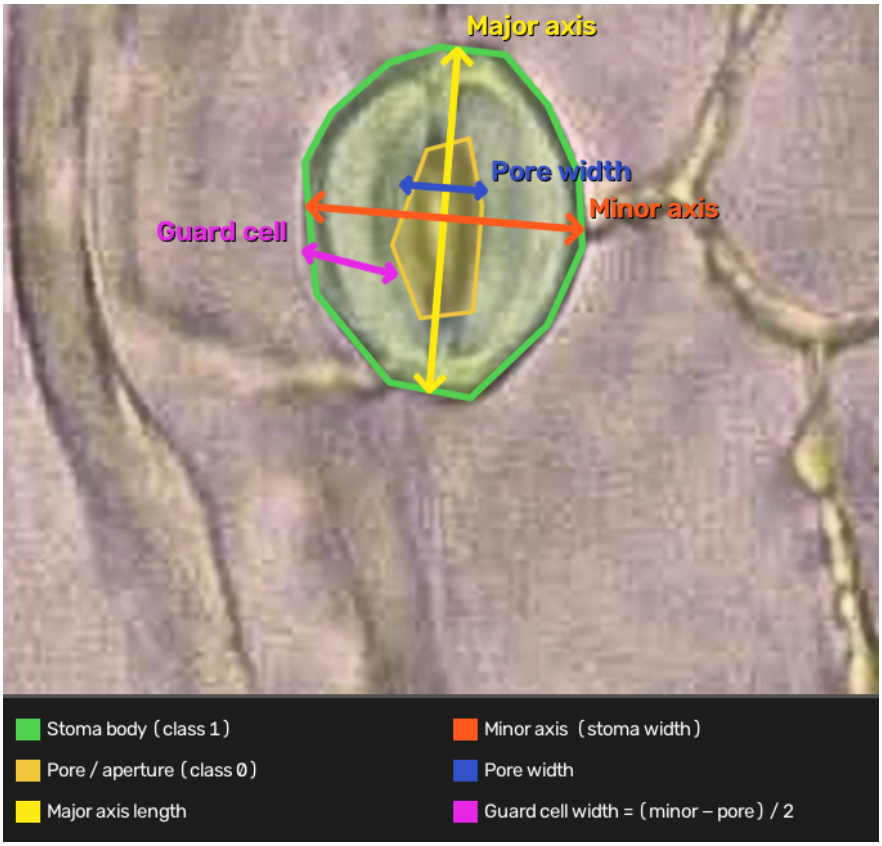


Figure 1: Measurement and trait extraction on a detected stoma from a microscope image. Green polygon: segmented whole stoma boundary (YOLO class 1); cyan polygon: pore/center boundary (YOLO class 0). Overlaid measurement vectors: Major axis (yellow, stoma length L), Minor axis (orange, stoma width W), Pore width / aperture (blue, w_a), and Guard-cell thickness (magenta bracket, w_{gc}). Stomatal density (D) is calculated as total detected stomata divided by imaged field area ($0.230\,\mathrm{mm}^2$).

Biophysical traits. - **Theoretical anatomical maximum conductance ($g_{s,\max}$)**: Computed via Franks & Farquhar (2001) and Sack & Buckley (2016): $g_{s,\max} = \frac{d}{v} \frac{D\, a_{\max}}{l + \frac{\pi}{2}\sqrt{a_{\max}/\pi}}$, where d is water vapor diffusivity in air ($2.49 \times 10^{-5}\,\mathrm{m}^2/\mathrm{s}$), v is molar volume ($0.02446\,\mathrm{m}^3/\mathrm{mol}$), and pore depth l is guard-cell thickness (w_{gc}). To prevent bias from variable momentary pore openness or drying artifacts during sampling, $a_{\max}$ uses standardized maximal geometry ($a_{\max} = \frac{\pi}{8} L_{\mathrm{pore}}^2$ with $L_{\mathrm{pore}} = 0.5\,L$). - **Stomatal Pore Area Index (SPI)**: $\mathrm{SPI} = D \times L^2$ (dimensionless). A pure structural proxy requiring zero assumptions regarding pore depth or aperture state.

Statistics. Each trait was analysed per microscope image with a linear mixed model, `trait ~ cytotype * leaf_side + (1 | population) + (1 | population:plant)`, accounting for repeated measures. Pairwise comparisons use Tukey-adjusted contrasts. In Figure 2, differences are denoted using **compact letter display (CLD)**.

Results

Table 1: Model-estimated marginal means (EMMs) per cytotype and leaf side ($g_{s,\max}$ in mol m^{-2} s^{-1}; SPI dimensionless).

	Abaxial				Adaxial			
Cytotype	Length (μm)	Density (mm^{-2})	SPI	$g_{s,\max}$	Length (μm)	Density (mm^{-2})	SPI	$g_{s,\max}$
ory_2x	18.6	294.6	0.099	0.83	19.1	240.8	0.083	0.69
nat_2x	18.8	252.5	0.089	0.73	19.2	205.5	0.072	0.59
nat_4x	20.6	234.8	0.097	0.73	21.4	208.8	0.089	0.67
neo_4x	23.9	161.9	0.090	0.60	24.7	135.0	0.076	0.51

Table 2: Selected pairwise differences (first minus second cytotype), Tukey-adjusted within leaf side. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, ns not significant.

	Abaxial				Adaxial			
Contrast	Length	Density	SPI	$g_{s,\max}$	Length	Density	SPI	$g_{s,\max}$
nat_4x - nat_2x	+1.8 ***	-17.7 ns	+0.008 ns	+0.01 ns	+2.2 ***	+3.3 ns	+0.017 ***	+0.08 ns
neo_4x - nat_2x	+5.1 ***	-90.6 ***	+0.001 ns	-0.13 ***	+5.5 ***	-70.6 ***	+0.004 ns	-0.08 **
neo_4x - nat_4x	+3.3 ***	-72.9 ***	-0.007 ns	-0.13 ***	+3.3 ***	-73.8 ***	-0.013 **	-0.15 ***
ory_2x - nat_2x	-0.2 ns	+42.0 ***	+0.010 *	+0.10 **	-0.1 ns	+35.3 **	+0.011 **	+0.10 **

Table 3: Variance decomposition across biological and experimental levels (% of total phenotypic variance explained). Marginal R^2_m captures fixed effects; conditional R^2_c captures total model explanatory power.

Source of variation	Length	Density	SPI	$g_{s,\max}$
Cytotype (Ploidy)	43.7%	16.8%	3.3%	7.2%
Leaf side	0.6%	3.2%	4.8%	4.1%
Cytotype × Side	0.0%	0.3%	0.4%	0.4%
Population of origin	18.6%	18.0%	8.2%	17.2%
Plant ID (within population)	26.3%	45.7%	54.1%	49.4%
Residual (image-to-image)	10.7%	16.0%	29.2%	21.7%
R^2_m Marginal (Fixed effects)	44.4%	20.3%	8.5%	11.7%
R^2_c Conditional (Full model)	89.3%	84.0%	70.8%	78.3%

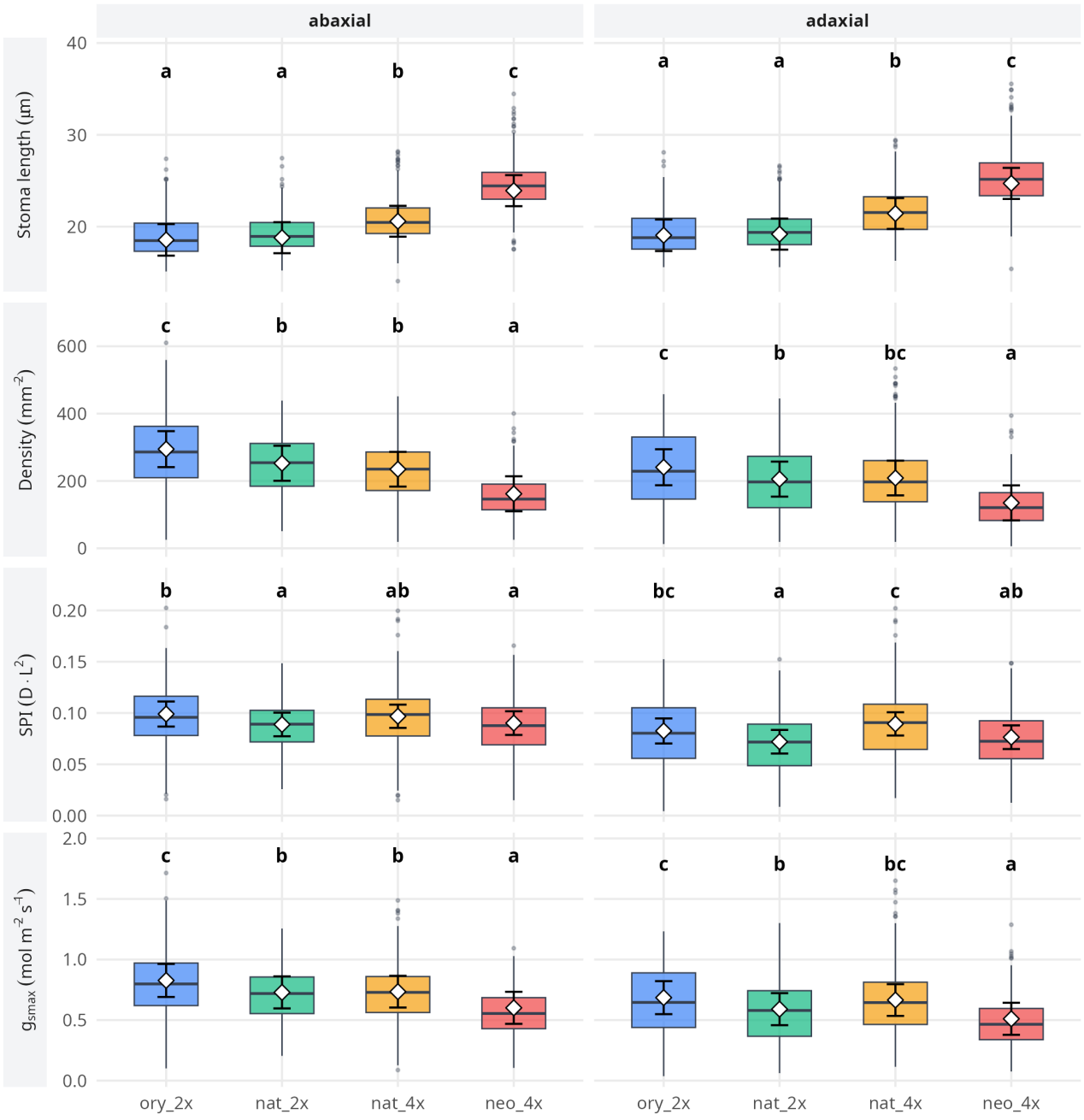


Figure 2: Distributions of stoma length, density, SPI, and anatomical conductance ($g_{s,\max}$) per microscope image across cytotypes and leaf sides. Boxes show image-level distributions; white diamonds with bars show model EMMs ± 95% CI. Letters above boxes display Tukey-adjusted compact letter display (CLD; $p < 0.05$) within each leaf side; cytotypes sharing a letter do not differ significantly.

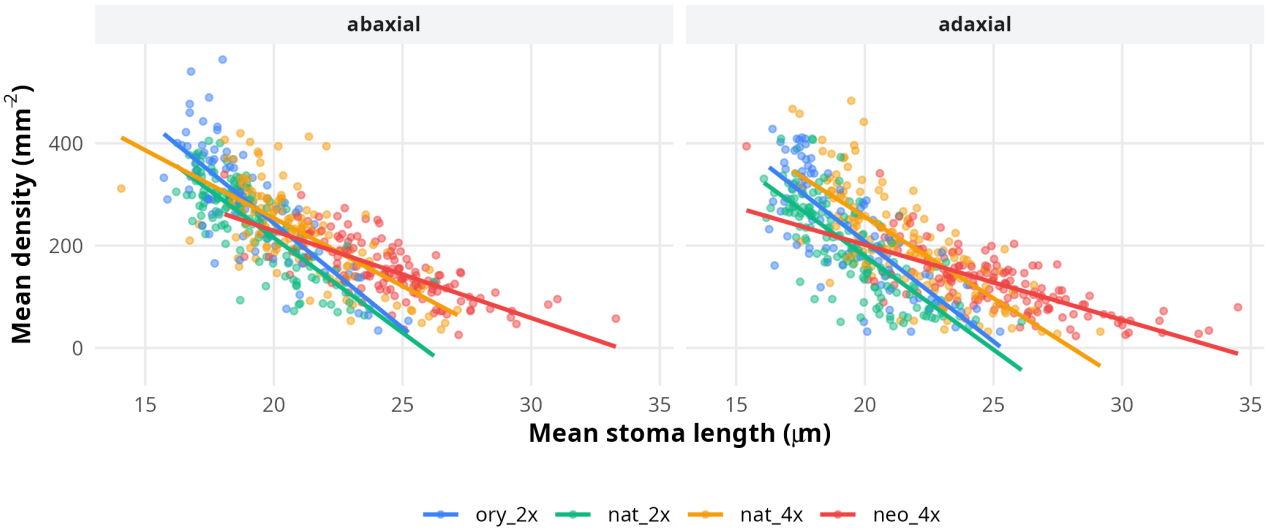


Figure 3: Size–density trade-off across cytotypes. Each point represents a single plant mean; solid lines are per-cytotype ordinary least squares fits.

Glossary and Technical Definitions

Microscope Image (Micrograph):

Simply an individual photograph taken through the microscope camera. Each image captures one field of view on a leaf surface imprint (covering $0.230\,\text{mm}^2$ of the epidermis). In this report, "3,253 images" or "3,253 micrographs" means 3,253 individual image files.

Population:

The geographic accession / field population of origin. The 5 populations included in the study are `AA012` and `AA025` (diploid origins), `AA225` and `AA310` (natural tetraploid origins), and `AA372` (mixed cytotype origin).

Cytotype:

The chromosomal ploidy classification of sampled individuals: - `nat_2x`: Natural diploid reference ($2n = 2x$). - `ory_2x`: Diploid control treated with oryzalin but retaining diploid ploidy. - `nat_4x`: Established natural tetraploid ($2n = 4x$). - `neo_4x`: Synthetic autotetraploid induced via mitotic spindle disruption with colchicine/oryzalin.

Stomata on Both Leaf Surfaces (Abaxial vs. Adaxial):

Leaves develop functional stomata on both the upper and lower epidermis surfaces rather than restricting them to the underside. **Abaxial** refers to the lower surface (typically shaded and carrying ~20–25% higher stomatal density); **adaxial** refers to the upper surface facing the sun.

Theoretical Anatomical Maximum Conductance ($g_{s,\max}$):

The physical structural ceiling for water vapor diffusion through the stomatal network (mol m^{-2} s^{-1}) following the biophysical diffusion model of Franks & Farquhar (2001) and Sack & Buckley (2016). Expressed per unit leaf area as:

$$g_{s,\max} = \frac{d}{v} \frac{D\, a_{\max}}{l + \frac{\pi}{2}\sqrt{a_{\max}/\pi}}$$

where d is diffusivity ($2.49 \times 10^{-5}\,\text{m}^2/\text{s}$), v is molar volume ($0.02446\,\text{m}^3/\text{mol}$), D is stomatal density, and pore depth l is guard-cell thickness (w_{gc}).

Pore Openness & Theoretical Maximum Geometry:

In physical imprints or peels, observed pore aperture fluctuates with momentary environmental conditions (light, VPD, time of sampling) and physical drying dynamics during imprint preparation. Calculating $g_{s,\max}$ directly from momentary aperture introduces sampling noise. Following Franks & Beerling (2009) and Sack & Buckley (2016), the anatomical maximum aperture is modeled from stoma geometry as $w_{\max} = 0.5\, L_{\text{pore}}$ (yielding $a_{\max} = \frac{\pi}{8} L_{\text{pore}}^2$). This standardizes $g_{s,\max}$ into a pure constitutional trait.

Stomatal Pore Area Index (SPI):

A dimensionless anatomical metric defined as $\text{SPI} = \text{Density} \times L_{\text{stoma}}^2$ (Sack et al. 2003). It quantifies the relative leaf surface area invested into the stomatal apparatus without requiring any assumptions regarding pore depth, 3D throat resistance, or aperture state.

Estimated Marginal Means (EMM):

Model-adjusted marginal means derived from linear mixed-effects models (`lmer`), accounting for unbalanced repeated sampling across individual plants (`plant_ID`) nested within populations.

Compact Letter Display (CLD):

A lettering scheme (Piepho 2004) summarizing all pairwise Tukey-adjusted hypothesis tests ($p < 0.05$). Treatments sharing at least one common letter are not statistically distinguishable, whereas groups with distinct letters differ significantly.